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# Epidemiology, Patterns of Care, and Mortality for Patients With Acute Respiratory Distress Syndrome in Intensive Care Units in 50 Countries

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Editorial page 759

Author Audio Interview at jama.com

Supplemental content at jama.com

CME Quiz at jamanetworkcme.com and CME Questions page 815

**IMPORTANCE** Limited information exists about the epidemiology, recognition, management, and outcomes of patients with the acute respiratory distress syndrome (ARDS).

**OBJECTIVES** To evaluate intensive care unit (ICU) incidence and outcome of ARDS and to assess clinician recognition, ventilation management, and use of adjuncts—for example prone positioning—in routine clinical practice for patients fulfilling the ARDS Berlin Definition.

**DESIGN, SETTING, AND PARTICIPANTS** The Large Observational Study to Understand the Global Impact of Severe Acute Respiratory Failure (LUNG SAFE) was an international, multicenter, prospective cohort study of patients undergoing invasive or noninvasive ventilation, conducted during 4 consecutive weeks in the winter of 2014 in a convenience sample of 459 ICUs from 50 countries across 5 continents.

**EXPOSURES** Acute respiratory distress syndrome.

**MAIN OUTCOMES AND MEASURES** The primary outcome was ICU incidence of ARDS. Secondary outcomes included assessment of clinician recognition of ARDS, the application of ventilatory management, the use of adjunctive interventions in routine clinical practice, and clinical outcomes from ARDS.

**RESULTS** Of 29 144 patients admitted to participating ICUs, 3022 (10.4%) fulfilled ARDS criteria. Of these, 2377 patients developed ARDS in the first 48 hours and whose respiratory failure was managed with invasive mechanical ventilation. The period prevalence of mild ARDS was 30.0% (95% CI, 28.2%-31.9%); of moderate ARDS, 46.6% (95% CI, 44.5%-48.6%); and of severe ARDS, 23.4% (95% CI, 21.7%-25.2%). ARDS represented 0.42 cases per ICU bed over 4 weeks and represented 10.4% (95% CI, 10.0%-10.7%) of ICU admissions and 23.4% of patients requiring mechanical ventilation. Clinical recognition of ARDS ranged from 51.3% (95% CI, 47.5%-55.0%) in mild to 78.5% (95% CI, 74.8%-81.8%) in severe ARDS. Less than two-thirds of patients with ARDS received a tidal volume 8 of mL/kg or less of predicted body weight. Plateau pressure was measured in 40.1% (95% CI, 38.2-42.1), whereas 82.6% (95% CI, 81.0%-84.1%) received a positive end-expository pressure (PEEP) of less than 12 cm $H_2O$. Prone positioning was used in 16.3% (95% CI, 13.7%-19.2%) of patients with severe ARDS. Clinician recognition of ARDS was associated with higher PEEP, greater use of neuromuscular blockade, and prone positioning. Hospital mortality was 34.9% (95% CI, 31.4%-38.5%) for those with mild, 40.3% (95% CI, 37.4%-43.3%) for those with moderate, and 46.1% (95% CI, 41.9%-50.4%) for those with severe ARDS.

**CONCLUSIONS AND RELEVANCE** Among ICUs in 50 countries, the period prevalence of ARDS was 10.4% of ICU admissions. This syndrome appeared to be underrecognized and undertreated and associated with a high mortality rate. These findings indicate the potential for improvement in the management of patients with ARDS.

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Acute respiratory distress syndrome (ARDS) is an acute inflammatory lung injury, associated with increased pulmonary vascular permeability, increased lung weight, and loss of aerated lung tissue.[1] Although prior epidemiologic studies have provided substantial insights into ARDS,[2-5] there remains limited information about the epidemiology, recognition, management, and outcomes of patients with the ARDS, especially in the era of the current Berlin Definition.[1] This definition was constructed empirically and validated using retrospective cohorts[1]; however, prospective studies of the Berlin Definition have been limited to small numbers of centers and patients.[6,7]

We set out to address some clinically important questions regarding ARDS. The current incidence and mortality of ARDS in a large international cohort is not known. Large regional differences have been suggested; for example, the incidence of ARDS in Europe[5] is reported to be 10-fold lower than in the United States.[4] A number of ventilatory interventions, such as lower tidal volumes,[8] higher positive end-expiratory pressure (PEEP),[9] and adjuncts such as prone positioning,[10] neuromuscular blockade,[11] and extracorporeal membrane oxygenation[12] for ARDS have been proposed. It is not clear how these interventions are applied in routine practice in the broader international context. Implementation of effective therapies may be limited by lack of recognition of ARDS by clinicians.[13,14] Understanding the factors associated with ARDS recognition and its effect on management could lead to effective interventions to improve care.

Therefore, we undertook the Large Observational Study to Understand the Global Impact of Severe Acute Respiratory Failure (LUNG SAFE) to determine the intensive care unit (ICU) epidemiology and outcomes from ARDS, assess clinician recognition of ARDS, and understand how clinicians use mechanical ventilation and adjunctive interventions in routine clinical practice.

## Methods

### Study Design

This study was an international, multicenter, prospective cohort study. The enrollment window consisted of 4 consecutive winter weeks (February-March 2014 in the Northern hemisphere and June-August 2014 in the Southern hemisphere), as selected by each ICU. We aimed to recruit a broadly representative sample of ICUs by public announcements by the European Society of Intensive Care Medicine, by national societies and networks endorsing the study, and by designated national coordinators (eAppendix 1 in the Supplement). The study ICUs represent a convenience sample of those that agreed to participate in the study and had enrolled at least 1 patient. Different ICUs from the same hospital were considered as separate centers; each ICU provided baseline data concerning its resources (eTable 1 in the Supplement). All participating ICUs obtained ethics committee approval and obtained either patient consent or ethics committee waiver of consent. We recruited physicians from each participating country as lead site investigators and national coordinators. Site investigators (eAppendix 2 in the Supplement) were also responsible for ensuring data integrity and validity, and were offered web-based training to enhance chest x-ray interpretation reliability as part of a substudy.

### Patients, Study Design, and Data Collection

All patients, including ICU transfers, admitted to an ICU within the 4-week enrollment window and receiving invasive or noninvasive ventilation were enrolled. Exclusion criteria were age younger than 16 years or inability to obtain informed consent, when required. Following enrollment, patients were evaluated daily for acute hypoxemic respiratory failure, defined as the concurrent presence of (1) ratio of arterial oxygen tension to inspired fraction of oxygen ($PaO_2/FIO_2$) of 300 mm Hg or less; (2) new pulmonary parenchymal abnormalities on chest x-ray or computed tomography; and (3) ventilatory support with continuous positive airway pressure (CPAP), expiratory positive airway pressure (EPAP), or positive end-expiratory pressure (PEEP) of 5 cm $H_2O$ or more

Day 1 was defined as the first day that acute hypoxemic respiratory failure criteria were satisfied, irrespective of ICU admission date. The case report form (eAppendix 3 in the Supplement) automatically prompted investigators to provide an expanded data set for days 1, 2, 3, 5, 7, 10, 14, 21, and 28 or at ICU discharge or death. All data were recorded at the same time, normally as close as possible to 10 AM each day. Patient outcomes included date of liberation from mechanical ventilation and vital status at ICU discharge and at either hospital discharge or at day 90, whichever occurred earlier.

### Quality Control

At the time of data entry, the site investigators were required to answer all queries raised by the case report form before they could electronically finalize a patient data set. Patient data sets that were not finalized were not included in the analysis (**Figure 1**). In addition, prior to analysis, all data were screened for potentially erroneous data and outliers. These data were verified or corrected by site investigators. We followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement guidelines for observational cohort studies.[15]

### Identification and Recognition of ARDS

The diagnosis of ARDS was made by a computer algorithm in the analysis phase of the study using the raw data that made up the various components of the Berlin ARDS Definition: (1) presence of acute hypoxemic respiratory failure criteria, (2) onset within 1 week of insult, or new (within 7 days) or worsening respiratory symptoms; (3) bilateral airspace disease on chest x-ray or computed tomography not fully explained by effusions, lobar or lung collapse, or nodules; and (4) cardiac failure not the primary cause of acute hypoxemic respiratory failure.

We assessed clinician recognition of ARDS at 2 time points. On day 1 of study entry, site investigators indicated the reasons for the patient's hypoxemia, with ARDS included as a potential cause. If the answer was "yes," ARDS was deemed to have been clinician-recognized on day 1. When patients exited the study, investigators were asked if the patient had ARDS at any stage during their ICU stay. ARDS was deemed to have



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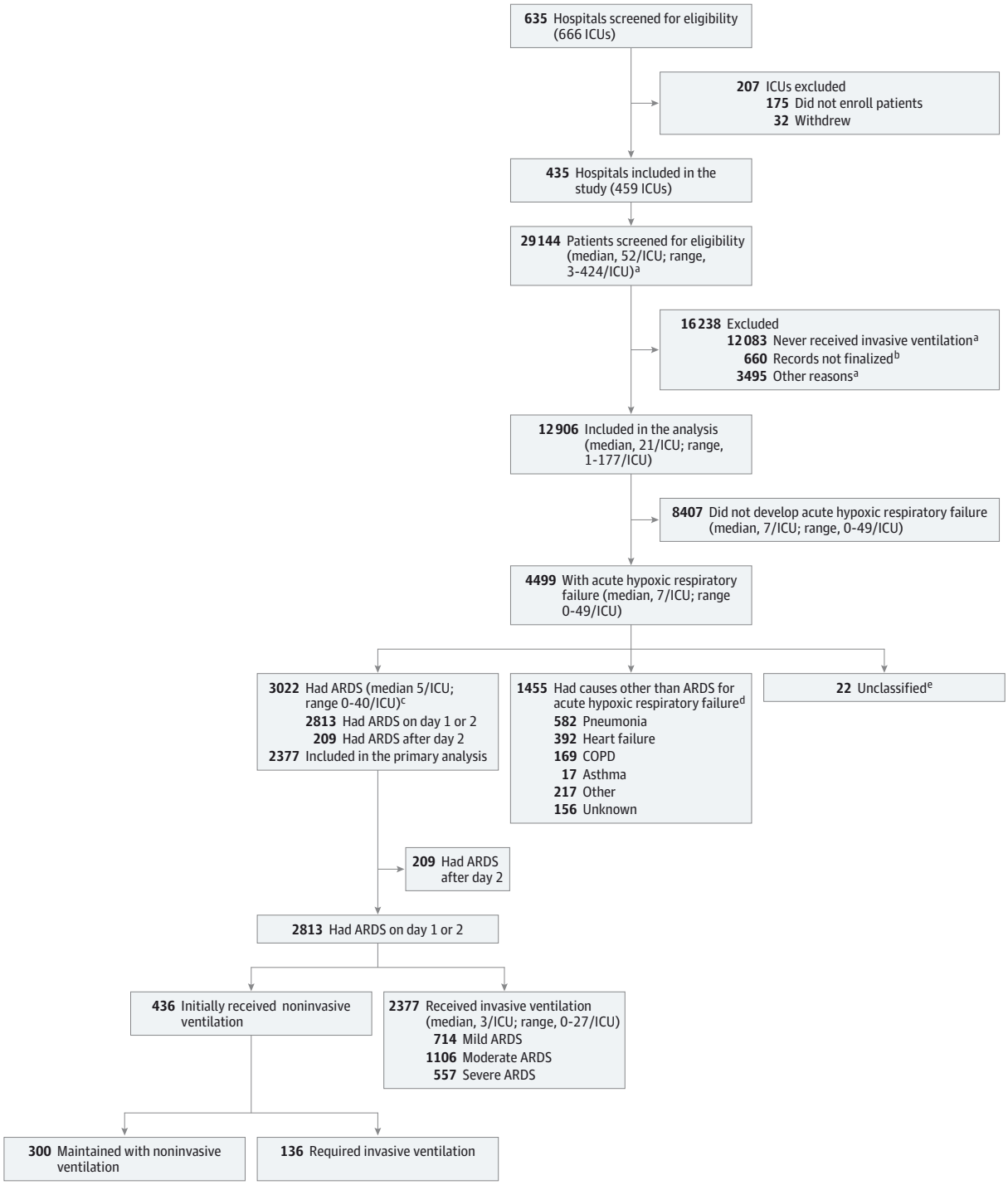
Figure 1. Flow of Patient Screening and Enrollment




[a] Projected from data provided by 360 intensive care units (ICUs [78%]). Data specifying other reasons were not collected during the study.

[b] Patient electronic case report forms that were not fully complete were excluded.

[c] Number included in the primary analysis.

[d] Patients could have more than one cause for acute hypoxic respiratory failure.

[e] For unclassified patients it was not possible to determine whether they fulfilled the criteria for acute respiratory distress syndrome (ARDS) due to incomplete data.

been clinician-recognized at any point if either question was answered positively. Although clinicians were offered participation in a substudy to evaluate a training module on chest x-ray diagnosis of ARDS, they were not specifically prompted with the Berlin criteria when answering the questions about ARDS diagnosis. Criteria for other diagnoses, such as chronic



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obstructive pulmonary disease, pneumonia, etc were left to clinician discretion.

### ARDS Severity and Mechanical Ventilation Parameters

Patients with ARDS undergoing invasive ventilation were categorized on the day of ARDS diagnosis based on their $PaO_2/FIO_2$ ratio into mild (200 <$PaO_2/FIO_2$ ≤300 mm Hg), moderate (100 <$PaO_2/FIO_2$ ≤200 mm Hg), and severe ($PaO_2/FIO_2$ <100 mm Hg) based on the Berlin Definition.[1] Given the lack of clarity in the Berlin Definition regarding the severity classification of patients managed with noninvasive ventilation, and the difficulty in comparing noninvasive ventilation settings to invasive modes, we excluded patients ventilated on noninvasive ventilation from the analyses pertaining to severity, ventilator management or outcome. To ensure a more homogenous data set, we restricted subsequent analyses to the large subset of patients (93.1%) fulfilling ARDS criteria on day 1 or 2 from onset of acute hypoxemic respiratory failure.

Invasive ventilator-free days were calculated as the number of days from weaning from invasive ventilation to day 28. Patients who died before weaning were considered to have a ventilator-free-day value of 0. Driving pressure was defined as plateau pressure ($P_{plat}$) minus PEEP.

Patients were considered to have no evidence for spontaneous ventilation when set and measured respiratory rates were equal.

### Calculation of Period Prevalence and Per-ICU-Bed ARDS Incidence

The period prevalence of patients with ARDS was calculated by dividing the number of patients fulfilling ARDS criteria by the total number of patients admitted to the ICU in the 28-day study period (ie, 29 160). The number of patients with ARDS per ICU bed over the 4-week study period was calculated as number of patients with ARDS/number of ICU beds available.

### ICU Enrollment and Statistical Analysis

The primary outcome was to determine the ICU incidence of ARDS. Secondary outcomes included assessment of clinician recognition of ARDS, the application of ventilatory management, the use of adjunctive interventions in routine clinical practice, and the outcomes from ARDS. We wished to enroll at least 1000 patients with ARDS. Assuming a 30% mortality, 300 deaths would allow us to evaluate at least 30 associated variables in multivariable models.[16] Prior epidemiological studies reported an ARDS incidence ranging between 2.2% and 19% of ICU patients.[2-5] Based on a conservative a priori estimate that 5% of ICU admissions would have ARDS and projecting that a medium-sized ICU admits 50 patients per month, we planned to enroll at least 500 ICUs worldwide.

Descriptive statistics included proportions for categorical and mean (standard deviation) or median (interquartile range [IQR]) for continuous variables. The amount of missing data was low, with the exception of plateau pressure $P_{plat}$ and arterial oxygen saturation ($Sa_{O_2}$), and is detailed in eTable 2 in the Supplement). No assumptions were made for missing data. Data were unadjusted unless specifically stated otherwise. Proportions were compared using $\chi^2$ or Fisher exact tests and continuous variables were compared using the *t* test or Wilcoxon rank-sum test, as appropriate. To evaluate variables associated with clinician recognition of ARDS, covariates determined a priori to be associated with ARDS recognition and covariates associated with ARDS recognition with $P < .20$ in bivariate analyses were entered into multivariable regression models with variable selection based on a stepwise backward elimination pro-

Table 1. Characteristics of Patients With Acute Respiratory Distress Syndrome

| Parameter | Value |
|---|---|
| No. of patients | |
| ARDS | 3022 |
| ARDS in first 48 h after AHRF | 2813 |
| No longer fulfill ARDS criteria after 24 h, No. (%) [95% CI] | 486 (17) [15.9-18.7] |
| Clinician recognition of ARDS, No. (%) [95% CI] | 1820 (60) [59-62.0] |
| Age, mean (95% CI)y | 61.5 (60.9-62.1) |
| Women, No. (%) | 1151 (38) |
| Height, mean (95% CI), cm | 168 (167.6-168.4) |
| Weight, mean (95% CI), kg | 78.0 (77-79) |
| Chronic disease, No. (%) | |
| COPD | 657 (21.7) |
| Diabetes | 657 (21.7) |
| Immunoincompetence | 365 (12.1) |
| Chronic cardiac failure | 314 (10.4) |
| Chronic renal failure | 306 (10.1) |
| Active neoplasm | 258 (8.5) |
| Hematological disease | 142 (4.7) |
| Risk factor for ARDS, No. (%)[a] | |
| Pneumonia | 1794 (59.4) |
| Extrapulmonary sepsis | 484 (16.0) |
| Aspiration | 430 (14.2) |
| Noncardiogenic shock | 226 (7.5) |
| Trauma | 127 (4.2) |
| Blood transfusion | 118 (3.9) |
| Pulmonary contusion | 97 (3.2) |
| Inhalation | 72 (2.3) |
| Drug overdose | 56 (1.9) |
| Pulmonary vasculitis | 41 (1.4) |
| Burn | 9 (0.3) |
| Drowning | 2 (0.1) |
| Other risk factor | 82 (2.7) |
| No risk factor | 252 (8.3) |
| Duration of invasive mechanical ventilation, median (IQR), d | 8 (4-16) |
| Duration of ICU stay, median (IQR), d | 10 (5-19) |
| ICU survival, No. (%) [95% CI] | 1994 (66.0) [64.3-67.7] |
| Duration of hospital stay, median (IQR), d | 17 (9-32) |
| Hospital survival, No. (%) [95% CI][b] | 1826 (60.4) [58.7-62.2] |

Abbreviations: AHRF, acute hypoxemic respiratory failure; ARDS, acute respiratory distress syndrome; COPD, chronic obstructive pulmonary disease; ICU, intensive care unit; IQR (interquartile range).

[a] Total is greater than 100%, because patients could have more than 1 risk factor.

[b] Data are missing for 10 patients.



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Table 2. Organizational and Patient Factors Associated With Clinician Recognition of ARDS in Invasively Ventilated Patients

| | ARDS Recognized, No./Total No. (%) | Absolute Difference (95% CI) | Bivariate OR (95% CI) | *P* Value[a] | Multivariable OR (95% CI) | *P* Value[b] |
|---|---|---|---|---|---|---|
| No. of patients/staff physician, for each additional patient | | −1.20 (−0.74 to −1.66)[c] | 0.960 (0.945 to 0.976) | <.001 | 0.959 (0.942 to 0.977) | <.001 |
| No. of patients/nurse, for each additional patient | | −0.34 (−0.55 to −0.13)[c] | 0.911 (0.860 to 0.957) | <.001 | 0.920 (0.870 to 0.968) | .002 |
| Age, per year | | −4.03 (−5.43 to −2.65)[c] | 0.985 (0.980 to 0.990) | <.001 | 0.987 (0.980 to 0.993) | <.001 |
| Predicted body weight per kg | | −1.27 (−2.18 to −0.36)[c] | 0.989 (0.980 to 0.997) | .006 | 0.984 (0.974 to 0.993) | <.001 |
| Nonpulmonary SOFA per point | | 0.81 (0.48 to 1.12)[c] | 1.054 (1.031 to 1.077) | <.001 | 1.057 (1.030 to 1.085) | <.001 |
| $Pa_{O_2}$:$FI_{O_2}$ ratio, per mm Hg | | −30.0 (−35.5 to −24.4)[c] | 0.993 (0.992 to 0.995) | <.001 | 0.993 (0.992 to 0.995) | <.001 |
| Medical or surgical Admission with trauma | | | | | | |
| No | 1477/2274 (65.0) | | 1 [Reference] | | 1 [Reference] | |
| Yes | 48/103 (46.6) | −18.4 (−28.7 to −8.0) | 0.471 (0.316 to 0.700) | <0.001 | 0.539 (0.334 to 0.868) | .011 |
| Neoplastic or immune or hematologic disease | | | | | | |
| No | 1173/1892 (62.0) | | 1 [Reference] | | 1 [Reference] | |
| Yes | 352/485 (72.6) | 10.6 (6.0 to 15.1) | 1.623 (1.305 to 2.027) | <.001 | 1.396 (1.079 to 1.816) | .012 |
| Pneumonia | | | | | | |
| No | 440/1002 (43.9) | | 1 [Reference] | | 1 [Reference] | |
| Yes | 963/1375 (70.0) | 26.1 (22.2 to 30.0) | 1.830 (1.544 to 2.170) | <.001 | 1.339 (1.073 to 1.670) | .01 |
| Pancreatitis | | | | | | |
| No | 844/2328 (36.3) | | 1 [Reference] | | 1 [Reference] | |
| Yes | 41/49 (83.7) | 47.4 (36.9 to 58.0) | 2.915 (1.436 to 6.733) | .006 | 3.506 (1.439 to 10.543) | .01 |
| ARDS risk factors | | | | | | |
| Yes | 1454/2187 (66.5) | | 1 [Reference] | | 1 [Reference] | |
| No | 71/190 (37.4) | −29.1 (−36.3 to −22.0) | 0.301 (0.220 to 0.408) | <.001 | 0.408 (0.280 to 0.591) | <.001 |
| With heart failure | | | | | | |
| No | 1347/2027 (66.5) | | 1 [Reference] | | 1 [Reference] | |
| Yes | 178/350 (50.9) | −15.6 (−21.2 to −10.0) | 0.522 (0.415 to 0.657) | <.001 | 0.496 (0.377 to 0.652) | <.001 |

Abbreviations. ARDS, acute respiratory distress syndrome; $Pa_{O_2}$/FIO2; partial pressure of oxygen to fraction of inspired oxygen; SOFA, Sequential Organ Failure Assessment.

[a] Bivariate analysis.

[b] All variables included in the multivariable analysis are reported in this Table.

[c] These values are the mean difference (95% CI).

cedure using *P* values. The association of clinician recognition with ventilatory management of ARDS was determined for tidal volume, PEEP, $P_{plat}$ measurement, and use of prone positioning and neuromuscular blockade in separate multivariable stepwise backward logistic or multiple linear regression models as appropriate. We did not perform any longitudinal data analyses. A Kaplan-Meier estimate of the cumulative probability of unassisted breathing and survival to day 28 was performed. Patients discharged from the hospital before day 28 were assumed alive at this time point. Statistical analyses were performed with R 3.2.3 (http://www.R-project.org). All *P* values were 2-sided, with *P* values <.05 considered statistically significant. The study protocol, case report form and full statistical analysis plan are included in eAppendix 3 in the Supplement.

## Results

### Participating ICUs and Patients Enrolled

Six hundred sixty-six ICUs registered for the study. Following data verification and elimination of nonrecruiting sites, 459 ICUs from 50 countries were included in the final analysis (eTable 1 and eTable 3 in the Supplement). Of the 29 144 patients admitted to these ICUs during the enrollment period, 13 566 patients receiving ventilatory support were enrolled. Complete data sets from 12 906 patients were analyzed (Figure 1). **Table 1** outlines their key characteristics.

### Characteristics of Patients Enrolled

Of 4499 patients with acute hypoxemic respiratory failure, 3022 (67.2%) fulfilled ARDS criteria during their ICU stay. Of these, 2813 (93.1%) developed ARDS at day 1 (n = 2665) or day 2 (n = 148), whereas 209 patients (6.9%) developed ARDS after day 2 of acute hypoxemic respiratory failure (Figure 1). The 436 patients (14.4%) with ARDS who received noninvasive ventilation were excluded from analyses regarding ARDS severity, mechanical ventilation settings, and outcome.

### ICU Incidence of ARDS

ARDS represented 10.4% (95% CI, 10.0%-10.7%) of total ICU admissions and 23.4% (95% CI, 21.7%-25.2%)of all patients requiring mechanical ventilation and constituted 0.42 cases/ICU bed over 4 weeks. There was some geographic variation, with Europe having an incidence of 0.48 cases/ICU bed over 4 weeks; North America, 0.46; South America, 0.31; Asia, 0.27; Africa, 0.32; and Oceania, 0.57 cases/ICU bed per 4 weeks.



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Table 3. Baseline Characteristics of Patients With Acute Respiratory Distress Syndrome Treated With Invasive Ventilation by Severity Category at Diagnosis

| Parameter | All (N = 2377) | Mild (n = 714) | Moderate (n = 1106) | Severe (n = 557) | *P* Value[a] |
|---|---|---|---|---|---|
| Age, median (IQR), y | 61 (61-62) | 61 (60-63) | 62 (62-63) | 57 (55-58) | <.001 |
| No longer meet ARDS criteria after 24 h, No. (%) [95% CI] | 486 (17.3) [15.9-18.7] | 190 (26.6) [23.4-30.0] | 152 (13.7) [11.8-15.9] | 71 (12.8) [10.1-15.8] | <.001 |
| **Severity of illness, mean (95% CI), SOFA score[b]** | | | | | |
| Day 1 | 10.1 (9.9-10.2) | 8.8 (8.6-9.1) | 10.2 (9.9-10.4) | 11.4 (11.1-11.8) | <.001 |
| Day 1 nonpulmonary[c] | 6.9 (6.7-7.0) | 6.7 (6.4-7.0) | 6.9 (6.7-7.1) | 7.0 (6.7-7.4) | .34 |
| Worst | 11.1 (10.9-11.3) | 10.3 (10.0-10.6) | 11.8 (11.5-12.0) | 13.0 (12.6-13.3) | <.001 |
| Worst nonpulmonary | 8.0 (7.8-8.2) | 8.0 (7.7-8.3) | 8.7 (8.4-8.9) | 9.0 (8.4-8.9) | <.001 |
| **Ventilator settings, first day of ARDS** | | | | | |
| $FIO_2$, mean (95% CI) | 0.65 (0.64-0.65) | 0.48 (0.47-0.50) | 0.62 (0.61-0.63) | 0.90 (0.88-0.91) | <.001 |
| Median (IQR) | 0.6 (0.45-0.85) | 0.4 (0.4-0.5) | 0.6 (0.5-0.7) | 1 (0.8-1) | |
| Set respiratory rate, mean (95% CI), 1/min | 18.6 (18.3-19.0) | 17.4 (16.9-17.8 | 18.4 (18.0-18.5) | 20.4 (19.2-21.6) | <.001 |
| Total respiratory rate, mean (95% CI), 1/min | 20.8 (21.5-21.2) | 19.5 (19.0-19.9) | 20.7 (20.3-21.1) | 22.7 (21.5-23.8) | <.001 |
| VT, mean (95% CI), mL/kg PBW | 7.6 (7.5-7.7) | 7.8 (7.6-7.9 | 7.6 (7.5-7.7) | 7.5 (7.3-7.6) | .02 |
| Control vent mode | 7.5 (7.4-7.6) | 7.6 (7.5-7.8) | 7.4 (7.3-7.6) | 7.4 (7.2-7.6) | .06 |
| Spontaneous vent mode | 7.9 (7.8-8.1) | 7.9 (7.7-8.2) | 8.0 (7.7-8.2) | 7.7 (7.4-8.1) | .55 |
| *P* value (control vs spont mode) | <.001 | .049 | <.001 | .053 | |
| Set PEEP, mean (95% CI), cm $H_2O$ | 8.4 (8.3-8.6) | 7.4 (7.2-7.6) | 8.3 (8.1-8.5) | 10.1 (9.8-10.4) | <.001 |
| Peak pressure, mean (95% CI), cm $H_2O$[d] | 27.0 (26.7-27.4) | 24.7 (24.1-25.4) | 26.9 (26.5-27.4) | 30.3 (29.6-30.9) | <.001 |
| Patients in whom $P_{PLAT}$ measured, No. (%) | | | | | |
| Among all invasively ventilated patients, No. (%) [95% CI] | 954 (40.1) [38.2-42.1] | 260 (36.4) [32.9-40.1] | 463 (41.9) [38.9-44.8] | 231 (41.5) [37.3-45.7] | .05 |
| Among patients with controlled ventilation, No. (%) [95% CI] | 756 (48.5) [46.0-51.0] | 198 (46.1) [41.3-51.0] | 363 (49.8) [46.1-53.5] | 195 (48.5) [43.5-53.5] | .49 |
| $P_{PLAT}$, mean (95% CI), cm $H_2O$[e] | 23.2 (22.6-23.7) | 20.5 (19.8-21.3) | 23.1 (22.6-23.7) | 26.2 (25.2-27.1) | <.001 |
| Standardized minute ventilation, mean (95% CI), l/min[f] | 10.8 (10.6-11.0) | 9.3 (9.1-9.6) | 10.7 (10.5-11.0) | 12.8 (12.3-13.3) | <.001 |
| Spontaneous ventilation, No. (%) [95% CI] | 723 (30.4 [8.6-32.3] | 260 (36.4) [32.9-40.0] | 336 (30.4) [29.7-35.3] | 127 (22.8) [19.3-26.5] | <.001 |
| **Gas exchange, first day of ARDS** | | | | | |
| $PaO_2/FIO_2$ ratio, mean (95% CI), mmHg | 161 (158-163) | 246 (244-248) | 149 (147-150) | 75 (74-77) | <.001 |
| $SpO_2$, mean (95% CI) | 95 (94-95) | 97 (97-98) | 95 (95-96) | 90 (89-91) | <.001 |
| Median (IQR) | 96 (93-98) | 98 (96-99) | 96 (94-98) | 92 (88-95) | |
| $PaCO_2$, mean (95% CI), mm Hg | 46.0 (45.4-46.6) | 41.5 (40.7-42.2) | 45.8 (44.9-46.6) | 52.2 (50.7-53.7) | <.001 |
| pH, mean (95% CI) | 7.33 (7.32-7.33) | 7.36 (7.36-7.37) | 7.33 (7.32-7.33) | 7.27 (7.26-7.29) | <.001 |

Abbreviations: ARDS, acute respiratory distress syndrome; IQR, interquartile range; PBW; predicted body weight; PEEP, positive end-expiratory pressure; $PaO_2/FIO_2$, partial pressure of oxygen to fraction of inspired oxygen; $P_{PLAT}$, plateau pressure; SOFA, Sequential Organ Failure Assessment; VT, tidal volume; $SpO_2$, peripheral arterial oxygen saturation.

[a] *P* value represents comparisons across the ARDS severity categories for each variable.

[b] For all SOFA scores for which data points were missing, this value was omitted and the denominator adjusted accordingly.

[c] The nonpulmonary SOFA score and the pulmonary component of the score was omitted and the denominator adjusted accordingly.

[d] For peak pressure measurements, patients receiving high-frequency oscillatory ventilation (HFOV) or extracorporeal membrane oxygenation (ECMO) were excluded.

[e] Plateau pressure values are limited to patients in whom this value was reported and in whom either an assist control mode was used or in whom a mode permitting spontaneous ventilation was used. The set and total respiratory rates were equal. Patients receiving HFOV or ECMO were also excluded.

[f] Standardized minute ventilation = minute ventilation × $PaCO_2$/40 mm Hg.

## Recognition of ARDS

ARDS was underdiagnosed, with 60.2% of all patients with ARDS being clinician-recognized. Clinician recognition of ARDS ranged from 51.3% (95% CI, 47.5%-55.0%) for mild ARDS to 78.5% (95% CI, 74.8%-81.8%) for severe ARDS (eTable 4 in the Supplement). Clinician recognition of ARDS at the time of fulfillment of ARDS criteria was 34.0% (95% CI, 32.0-36.0), suggesting that diagnosis of ARDS was frequently delayed.

A multivariable analysis including variables from the bivariable analyses (eTable 5 in the Supplement), revealed several patient and organizational factors associated with clinician recognition of ARDS. Higher nurse-to-patient ratios, higher physician-to-patient ratios, younger patient age and a lower $PaO_2/FIO_2$ ratio, and the presence of pneumonia or pancreatitis were factors independently associated with higher probability of clinician recognition (**Table 2**). Absence of a risk factor and presence of concomitant



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Table 4. Use of Adjunctive and Other Optimization Measures in Invasively Ventilated Patients With Acute Respiratory Distress Syndrome[a]

| | Patients of No. (%) [95% CI] | | | | |
|---|---|---|---|---|---|
| | All (n = 2377) | Mild[a] (n = 498) | Moderate[a] (n = 1150) | Severe[a] (n = 729) | *P* Value[b] |
| Neuromuscular blockade | 516 (21.7) [20.1-23.4] | 34 (6.8) [4.8-9.4] | 208 (18.1) [15.9-20.4 ] | 274 (37.8) [34.1-41.2] | <.001 |
| Recruitment maneuvers | 496 (20.9) [19.2-22.6] | 58 (11.7) [9.0-14.8] | 200 (17.4) [15.2-19.7] | 238 (32.7) [29.3-36.2] | <.001 |
| Prone positioning | 187 (7.9) [6.8-9.0] | 5 (1.0) [0.3-2.3] | 63 (5.5) [4.2-7.0] | 119 (16.3) [13.7-19.2] | <.001 |
| ECMO | 76 (3.2) [2.5-4.0] | 1 (0.2) [0.05-1.2] | 27 (2.4) [1.6-3.4] | 48 (6.6) [4.9-8.6] | <.001 |
| Inhaled vasodilators | 182 (7.7) [6.6-8.8] | 17 (3.4) [02.0-5.4] | 70 (6.1) [4.8-7.6] | 95 (13.0) [10.7-15.7] | <.001 |
| HFOV | 28 (1.2) [0.8-1.7] | 3 (0.6) [0.1-1.7] | 14 (1.2) [0.7-2.0] | 11 (1.5) [0.8-2.7] | .347 |
| None of the above | 1431 (60.2) [58.2-62.2] | 397 (79.7) [75.9-83.2] | 750 (65.2) [62.4-68.0] | 284 (39.0) [35.4-42.6] | <.001 |
| Esophageal pressure catheter | 19 (0.8) [0.04-1.4] | 2 (0.4) [0.04-1.4] | 8 (0.7) [0.3-1.3] | 9 (1.2) [0.6-2.3] | .233 |
| Tracheostomy | 309 (13.0) [11.6-14.4] | 48 (9.6) [7.1-12.6] | 155 (13.5) [11.6-15.6] | 106 (14.5) [12.1-17.3] | .034 |
| High-dose corticosteroids[c] | 425 (17.9) [16.4-19.5] | 61 (12.3) [9.5-15.5] | 194 (16.9) [14.7-19.2] | 170 (23.3) [20.3-26.6] | <.001 |
| Pulmonary artery catheter | 107 (4.5) [3.7-5.4] | 9 (1.8) [0.8-3.4] | 53 (4.6) [3.4-6.0] | 45 (6.2) [4.5-8.2] | .001 |

Abbreviations: ARDS, acute respiratory distress syndrome; ECMO, extracorporeal membrane oxygenation; HFOV, high-frequency oscillatory ventilation; PEEP, positive end-expiratory pressure.

[a] For this analysis, ARDS severity was defined based on the patients' worst severity category over the course of their ICU stay in patients who developed ARDS on day 1 or 2.

[b] *P* value represents comparisons across the ARDS severity categories for each variable.

[c] High-dose corticosteroids was defined as doses that were equal to or greater than the equivalent of 1 mg/kg of methylprednisolone.

Table 5. Outcome of Invasively Ventilated Patients by Acute Respiratory Distress Syndrome Severity at Diagnosis

| Parameter | All (n = 2377) | Mild (n = 714) | Moderate (n = 1106) | Severe (n = 557) | *P* Value[a] |
|---|---|---|---|---|---|
| Progression of ARDS severity, No (%) [95% CI][b] | | | | | |
| Progression to moderate[c] | | 184 (25.8) [22.6-29.1] | N/A | N/A | |
| Progression to severe[c] | | 32 (4.5) [3.1-6.3] | 140 (12.7) [10.8-14.8] | N/A | |
| Death in the 1st wk without category change | | 63 (8.8) [6.8-11.1] | 126 (11.4) [9.6-13.4] | 117 (21.0) [17.7-24.6] | |
| Invasive ventilation-free days to day 28, median (IQR), d[d] | 10 (0-22) | 16 (0-24) | 11 (0-21) | 0 (0-18) | <.001 |
| Duration of invasive ventilation, median (IQR), d | | | | | |
| All patients | 8 (4-15) | 7 (3-14) | 8 (4-16) | 9 (4-16) | .04 |
| Surviving patients | 8 (4-15) | 6 (3-13) | 8 (4-15) | 11 (6-18) | <.001 |
| ICU length of stay, median (IQR), d | | | | | |
| All patients | 10 (5-20) | 10 (5-19) | 11 (6-20) | 11 (5-19) | .39 |
| Surviving patients | 11 (7-21) | 10 (6-19) | 12 (7-21) | 14 (7-23) | .03 |
| ICU mortality, No. (%) [95% CI] | 838 (35.3) [33.3-37.2] | 212 (29.7) [26.4-33.2] | 387 (35.0) [32.2-37.9] | 239 (42.9) [38.8-47.1] | <.001 |
| Day 28 mortality, No. (%) [95% CI] | 828 (34.8) [32.9-36.8] | 211 (29.6) [26.2-33.0] | 389 (35.2) [32.4-38.1] | 228 (40.9) [36.8-45.1] | <.001 |
| Hospital length of stay, median (IQR), d | | | | | |
| All patients | 17 (8-33) | 18 (10-33) | 17 (8-33) | 16 (6-31) | .22 |
| Surviving patients | 23 (14-40) | 23 (14-40) | 22 (13-40) | 26 (14-43) | .41 |
| Hospital mortality, No. (%) [(95% CI] | 952 (40.0) [38.1-42.1] | 249 (34.9) [31.4-38.5] | 446 (40.3) [37.4-43.3] | 257 (46.1) [41.9-50.4] | <.001 |

Abbreviations: ARDS, acute respiratory distress syndrome, ICU, intensive care unit; IQR, interquartile range.

[a] *P* value represents comparisons across the ARDS severity categories for each variable.

[b] Initial ARDS severity determined from worst partial pressure of oxygen to fraction of inspired oxygen ratio within first 24 hours following ARDS diagnosis.

[c] Most severe is calculated for time period up to day 7 postdiagnosis of ARDS. Analysis was limited to the first 7 days due to the less frequent sampling after that day.

[d] In patients in whom death occurs while receiving invasive mechanical ventilation, invasive ventilation-free days are counted as 0.



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cardiac failure were associated with reduced likelihood of clinician recognition of ARDS (Table 2). The mean tidal volume was 7.5 mL/kg (95% CI, 7.4-7.6 mL/kg) of predicted body weight (PBW) among patients whose physicians recognized ARDS, marginally lower than that of 7.7 mL/kg (95% CI, 7.6-7.9 mL/kg) in patients whose ARDS was not recognized ($P$ = .01). The mean PEEP level was 8.9 cm $H_2O$ (95% CI, 8.8-9.1 cm $H_2O$) in patients whose ARDS was recognized, higher than that of 7.5 cm $H_2O$ (95% CI, 7.3-7.7 cm $H_2O$) in patients whose ARDS was not recognized ($P$ < .001). Physicians who recognized ARDS used adjunctive treatments more than physicians who did not (43.9% vs 21.7%, $P$ < .001; eTable 4 in the Supplement). After adjusting for potentially confounding variables, there was no statistically significant association between clinician-recognized ARDS and tidal volumes (eTable 6 in the Supplement) or $P_{plat}$ recording (eTable 7 in the Supplement). In contrast, clinician recognition of ARDS was statistically associated with the use of higher levels of PEEP, and greater use of prone positioning and neuromuscular blockade (eTables 8-10 in the Supplement).

## ARDS Severity

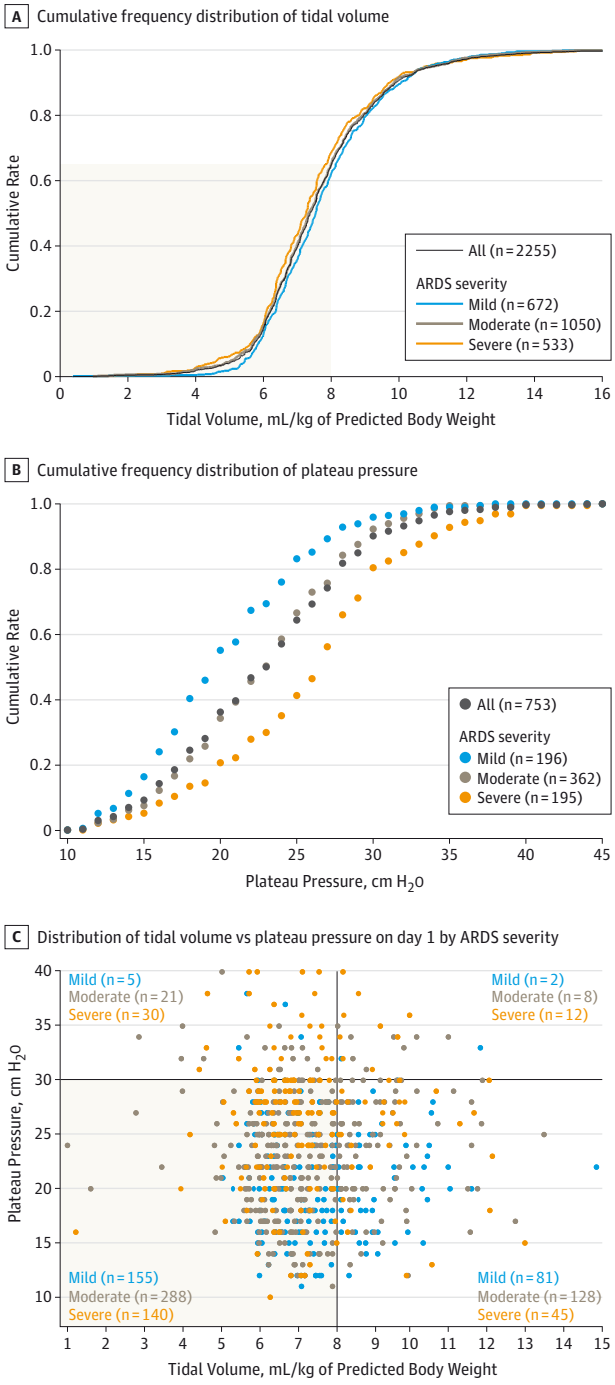
A total of 2377 patients developed ARDS in the first 48 hours of acute hypoxemic respiratory failure and received invasive mechanical ventilation. The period prevalence of mild ARDS was 30.0% (95% CI, 28.2%-31.9%); moderate, 46.6% (95% CI, 44.5%-48.6%); and severe, 23.4% (95% CI, 21.7%-25.2%) (Figure 1). Ventilator management differed among the ARDS severity groups, while the use of adjunctive measures increased and mortality was higher with greater ARDS severity (**Table 3**, **Table 4**, and **Table 5**). At diagnosis, increasing ARDS severity was paralleled by worsening Sequential Organ Failure Assessment (SOFA) scores, which was largely accounted for by the pulmonary component. The nonpulmonary component of the SOFA score was higher in patients with an increased ARDS severity category (Table 3). The $PaCO_2$ increased and pH decreased in patients with increased ARDS severity category (Table 3, eFigure 1A-B in the Supplement). Three hundred sixteen patients (13.3%) with ARDS had a $PaCO_2$ of 60 mm Hg or higher. However, the extent and severity of hypercapnia was relatively modest, even in severe ARDS.

## Mechanical Ventilation in ARDS

Ventilator management varied with ARDS severity (Table 3). However, the decrease in tidal volume and increase in PEEP, from mild to moderate to severe ARDS, while statistically significant, was clinically modest (Table 3). In patients with ARDS 35.1% (95% CI, 33.1%-37.1%) received a tidal volume of more than 8 mL/kg PBW (**Figure 2**A and eFigure 1C in the Supplement), while 82.6% (95% CI, 81.0%-84.1%) received a PEEP of less than 12 cm $H_2O$.

The distribution of $P_{plat}$ differed significantly with ARDS severity (Figures 2B and eFigure 1D in the Supplement). $P_{plat}$ was measured in 40.1% (95% CI, 46.0%-51.0%) of patients, irrespective of ARDS severity. This rose to 48.5% (95% CI, 46.0%-51.0%) of patients in whom there was no evidence for

**Figure 2. Ventilation Parameters in Patients With ARDS**




A, Cumulative frequency distribution of tidal volume was similar in patients in each severity category, with 65% of patients with acute respiratory distress syndrome (ARDS) receiving a tidal volume of 8 mL/kg of predicted body weight or less. B, In contrast, a right shift of the cumulative frequency distribution curves of plateau pressures was seen for increasing ARDS severity category, with plateau pressure of more than 30 cm $H_2O$ in 8.5% of patients for which these data are available. C, Represents the distribution of day-1 tidal volume vs plateau pressure for each patient for which these data are available. Two-thirds of the patients fell within the limits for *protective ventilation*, defined as plateau pressure less than or equal to 30 cm $H_2O$ and tidal volume of less than or equal to 8 mL/kg of predicted body weight. Data refer to the first day of ARDS.



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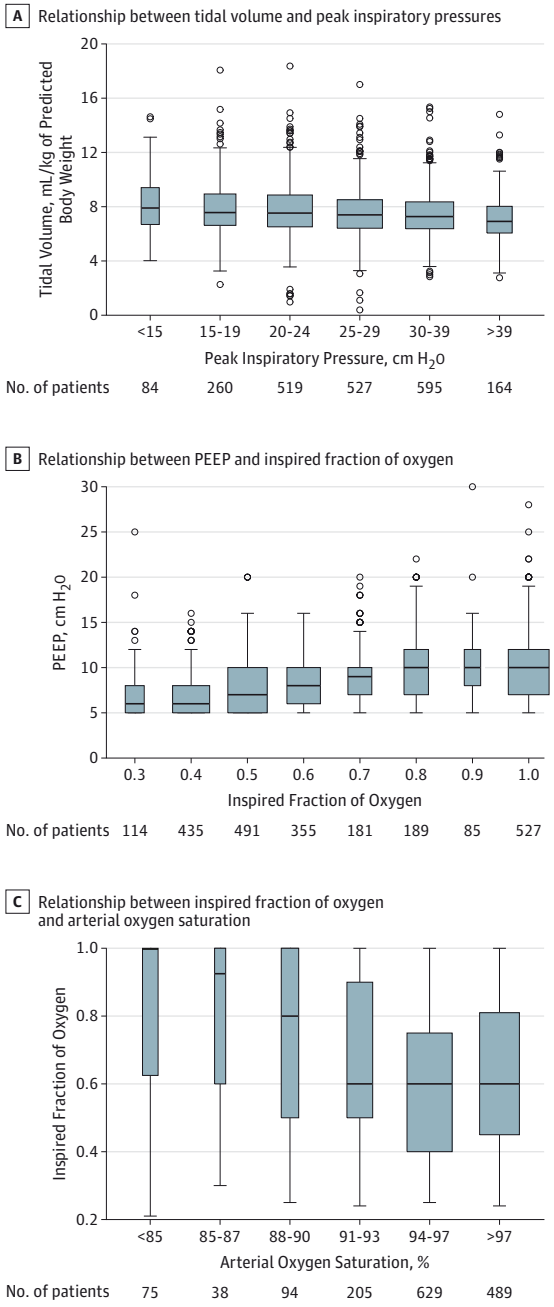
Figure 3. Mechanical Ventilation Settings in Early Acute Respiratory Distress Syndrome




A, Tidal volume remained relatively constant across the range of peak inspiratory pressures. B, Positive end-expiratory pressure (PEEP) progressively increased in patients requiring higher inspired fraction of oxygen ($FIO_2$). C, There was a stepwise increase in $FIO_2$ at lower arterial oxygen saturations, with $FIO_2$ steeply increasing at atrial oxygen saturation ($SaO_2$) values lower than 91%. Data refer to the first day of ARDS.

For each box plot, the middle line represents the median, the lower hinge represents the first quartile, the upper hinge represents the third quartile, the whiskers extend to 1.5 times interquartile range, and the outliers are values outside the whiskers' range. The boxes are drawn with widths proportional to the square root of the number of observations in the groups. The numbers below each box plot represent the total number of patients in each group.

spontaneous ventilation. Two-thirds of patients in whom $P_{plat}$ was reported received *protective mechanical ventilation* as defined by a tidal volume of 8 mL/kg of PBW or less and a $P_{plat}$ of 30 cm $H_2O$ or less (Figure 2C). In patients in whom $P_{plat}$ was measured, 91.9% (95% CI, 88.1%-94.9%) of those receiving a tidal volume of more than 8 mL/kg PBW had a $P_{plat}$ of 30 cm $H_2O$ or less (Figure 2C). Less than 3% of patients received a tidal volume of more than 8 mL/kg and had a $P_{plat}$ pressure of more than 30 cm $H_2O$ (Figure 2C).

There was no relationship between tidal volume and either peak inspiratory pressure, $P_{plat}$ or lung compliance (**Figure 3**A and eFigure 2 in the Supplement). Tidal volume was significantly higher in patients in a spontaneous breathing mode (7.5; 95% CI, 7.4-7.6 vs 7.9; 95% CI, 7.8-8.1 mL/kg PBW, $P < .001$; Table 3).

Positive end-expiratory pressure levels were relatively low (Table 3) and were higher in patients with higher peak inspiratory pressure and higher $P_{plat}$. In addition, no relationship was found between PEEP and the $PaO_2/FIO_2$ ratio, $FIO_2$ (Figure 3B) or lung compliance (eFigure 2 in the Supplement). In contrast, there was an inverse relationship between $FIO_2$ and $SpO_2$, suggesting that clinicians used $FIO_2$ to treat hypoxemia (Figure 3C).

## Use of Adjunctive Measures

The use of adjunctive treatments in patients with ARDS on day 1 or 2 was relatively low but increased with ARDS severity (Table 4). Continuous neuromuscular blocking agents, high-dose steroids, and recruitment maneuvers were the most frequently used adjuncts. In patients with severe ARDS, continuous neuromuscular blockade was used in 37.8% (95% CI, 34.1%-41.2%), prone position in 16.3% (95% CI, 13.7%-19.2%), and recruitment maneuvers in 32.7% (95% CI, 29.3%-36.2%).

## ARDS Outcomes

Severity of ARDS worsened in 356 (19.6%, 95% CI, 17.8%-21.5%) patients with mild or moderate ARDS (Table 5). There was a decreased likelihood of unassisted breathing (**Figure 4**A) and survival (Figure 4B) at day 28 with increasing severity. Overall, unadjusted ICU and hospital mortality from ARDS were 35.3% (95% CI, 33.3%-37.2%) and 40.0% (95% CI, 38.1%-42.1%), respectively (Figure 4 and Table 5). The number of ventilator-free days decreased (eFigure 3 in the Supplement), and the length of ICU—but not hospital—stay, increased with greater ARDS severity category. Both ICU and hospital survival decreased with increased ARDS severity (Table 5). Patients with a driving pressure (ie, $P_{plat}$-PEEP) of more than 14 cm $H_2O$ on day 1 had a worse outcome (Figure 4C). There was a direct relationship between both plateau and driving pressure quintile and mortality rate (**Figure 5**).

## Discussion

In this prospective study carried out in 459 ICUs in 50 countries in 5 continents, ARDS appeared to represent an impor-



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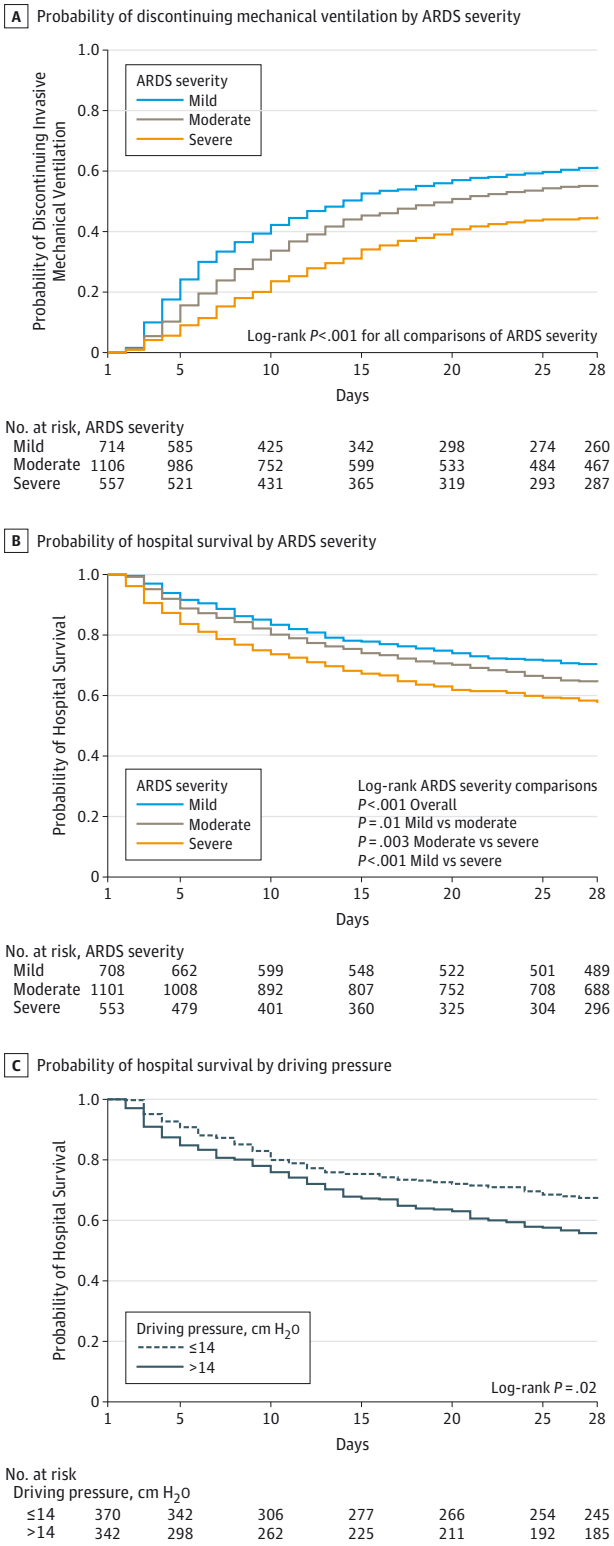
tant public health problem globally, with some geographic variation and with a very high mortality of approximately 40%. A major finding was the underrecognition of ARDS by clinicians, the low use of contemporary ventilatory strategies and adjuncts, and the limited effect of physician diagnosis of ARDS on treatment decisions. These findings indicate the potential for improvement in management of patients with ARDS.

In this study, the geographic variation in ARDS incidence ranged from 0.27 to 0.57 cases per ICU bed per 4 weeks and prcentage of ICU admissions. Because we could not estimate the population served by the ICUs in this study, we could not calculate population incidence for ARDS; therefore, relatively little can be inferred about the burden of ARDS in participating countries. The nearly 2-fold variation in ICU incidence in this study and the known variation in ICU resources internationally may well explain the variability in ARDS studies that involved specific geographic populations,[5] with the highest estimates in the United States[4,17] and Australia.[18,19] Our ICU incidence data are concordant with other estimates using similar approaches that have generated reliable population incidence data.[20]

These results suggest that ARDS continues to be underrecognized by clinicians in the era of the Berlin Definition, similar to previous findings using the American-European consensus conference (AECC) definition.[14,21-23] A key feature of our study design was that data were collected for each component of the Berlin Definition in all patients with hypoxemia breathing with the aid of a ventilator, which allowed us to identify patients with ARDS from the raw data. We chose this approach to enable a more robust evaluation of the incidence, as well to assess clinician recognition of ARDS. The rate of clinician recognition of ARDS was low, with 40% of all cases not being diagnosed. Clinician recognition rates increased with increasing disease severity but was still less than 80% in severe ARDS. Independent factors contributing to clinician recognition were younger patient age, lower predicted body weight, the presence of extrapulmonary sepsis or pancreatitis, and greater disease severity. Conversely, the absence of a risk factor for ARDS was associated with underrecognition of ARDS. Lower numbers of nurses and physicians per ICU patient were both associated with reduced clinician recognition of ARDS. It is possible that the way in which the data were collected contributed, in part, to clinician underrecognition of ARDS. Specifically, it is possible that the ICU clinician knew that the patient had ARDS, but this was not made known to the site investigators or reported in the patient chart. However, not indicating the diagnosis of ARDS in the chart constitutes a form of underrecognition. In addition, that the study had an explicit focus on ARDS, that all participants were offered online training on ARDS diagnosis, and that the case report form asked at 2 separate points in the study if the patient had ARDS, make this possibility less likely.

It is unclear whether clinician recognition of ARDS affects outcome because recognition may be only one of a number of barriers to the use of ventilatory and adjunctive

**Figure 4. Outcome From Acute Respiratory Distress Syndrome**




A, There was a lower likelihood of unassisted breathing with increasing severity of acute respiratory distress syndrome. B, There was a lower likelihood of survival to day 28 with increasing severity of acute respiratory distress syndrome (ARDS) at day 1. C, Patients with a driving pressure of greater than 14 cm $H_2O$ on day 1 of ARDS criteria had a higher mortality.



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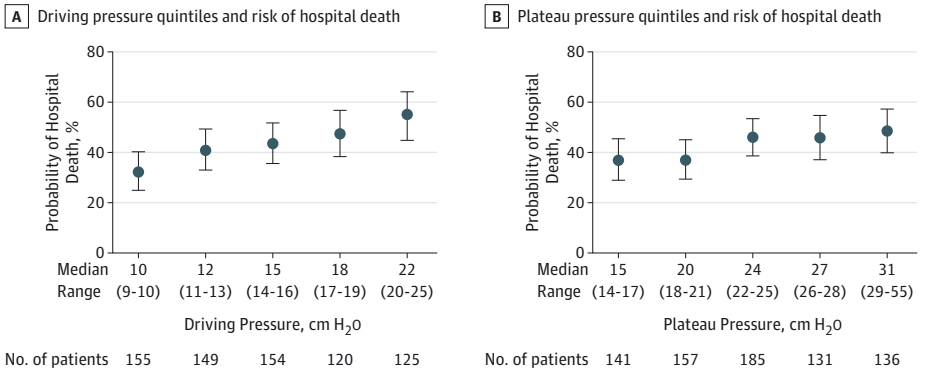
Figure 5. Driving Pressure and Plateau Pressure and Outcome From ARDS




A, Mortality increased with increasing quintiles of driving pressure on first day of ARDS. B, The relationship between mortality and quintiles of plateau pressure is provided for comparison. Error bars indicate 95% confidence interval.

treatment strategies, while the sickest patients are more frequently diagnosed.[14,24] After adjusting for potential confounders, clinician diagnosis of ARDS was not independently associated with the use of lower tidal volume. Conversely, clinician diagnosis of ARDS was significantly associated with the use of higher PEEP, prone positioning, and neuromuscular blockade. Although the reasons for this are unclear, clinicians do not appear influenced by the presence or absence of ARDS for setting tidal volume and may be motivated by other factors (eg, perceived comfort, pH, $PaCO_2$, etc).

Our data appear to demonstrate the predictive validity of the Berlin Definition, and are consistent with a recent observational study.[7] Increasing ARDS severity was associated with longer ICU stay, more days of invasive ventilation, longer hospital stays, and higher mortality. Patients with severe ARDS were younger, had fewer comorbidities but had a significantly worse outcome. The proportion of patients in each severity category was similar to that determined in retrospective analyses.[1]

ARDS appears to be undertreated in terms of the use of optimal, proven, or recommended approaches to mechanical ventilation and regarding the use of some adjunctive measures. Plateau pressure was reported in only 40.1% of all patients with ARDS, which increased to 48.5% of patients in whom there was no evidence for spontaneous ventilation. Although it is possible that patients in whom plateau pressure was measured were ventilated differently, this did not appear to be the case, at least in terms of tidal volume. We found no evidence to suggest that lower tidal volumes or higher PEEP were used in patients with a less compliant respiratory system or greater ARDS severity as reported in prior studies.[22] Low tidal volume ventilation was the most frequently used intervention, but more than one-third of all patients with ARDS received a tidal volume of more than 8 mL/kg of PBW, and approximately 60% received a tidal volume of more than 7 mL/kg of PBW. This finding is consistent with recent nonprotocolized RCTs in which patients received larger tidal volumes than expected.[12,25] In our study, PEEP was relatively low and constant across the spectrum of ARDS severity, with more than 80% of patients with ARDS receiving PEEP of 12 cm $H_2O$ or less. Hypoxemia appeared to be treated predominantly by increasing $FIO_2$. High levels of permissive hypercapnia were infrequent. Adjunctive measures were used infrequently; this appeared to be the case for less expensive interventions such as prone positioning and neuromuscular blockade, as well as for expensive and invasive technologies such as extracorporeal membrane oxygenation. It is possible that the relatively low use of adjunctive measures such as neuromuscular blockade or prone positioning reflects ongoing uncertainty about the quality of evidence supporting these interventions.

ARDS continues to have a high mortality, despite advances in supportive care. There was a significant increase in mortality with each increase in ARDS severity category. Overall, 40% of patients with ARDS died in the hospital. Although detailed analyses of the factors contributing to outcome are beyond the scope of this article, we also confirmed a recent report[26] suggesting that higher driving pressure is associated with increased risk of death; albeit, our data should be interpreted cautiously as $P_{plat}$ was available in a minority of patients.

This study has a number of limitations. Our focus on winter months, while allowing us to examine the burden of ARDS during the same season across the globe, may overstate ICU incidence figures for ARDS, due to specific diseases such as influenza.[27] In addition, despite enrolling a large number of ICUs from around the world, our convenience sample may be prone to selection biases that may limit generalizability; therefore, we are unable to calculate population-based incidence figures for ARDS. Similar to other epidemiological studies, we did not have access to the source data for the patients in the enrolling ICUs, so it is possible that not all patients with ARDS in participating centers were enrolled. However, enrollment of patients with ARDS from participating ICUs met expectations based on their recorded 2013 admission rates, while data from lower



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recruiting ICUs was not different from that from higher enrolling ICUs, suggesting the absence of reporting biases. To ensure data quality, we instituted a robust data quality-control program in which all centers were requested to verify data that appeared inconsistent or erroneous. Although chest x-ray interpretation was performed by on-site clinicians, which potentially increased variability, we attempted to standardize interpretation by offering all the investigators web-based training. Another limitation is the lack of data collection concerning the use of conservative fluid strategy. Lastly, our assumption that patients discharged from the hospital before day 28 were alive at that time point is a further limitation.

## Conclusions

Among ICUs in 50 countries, the period prevalence of ARDS was 10.4% of ICU admissions. This syndrome appeared to be underrecognized, undertreated, and associated with a high mortality rate. These findings indicate the potential for improvement in management of patients with ARDS.

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**Correction:** This article was corrected online July 19, 2016, for a language error in the Discussion section and for an incorrect list in the Supplement.

**Author Contributions:** Dr Pham and Dr Bellani had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.
*Study concept and design:* Bellani, Laffey, Pham, Fan, Brochard, Esteban, Gattinoni, Ranieri, Rubenfeld, Thompson, Wrigge, Slutsky, Pesenti.
*Acquisition, analysis, or interpretation of data:* Bellani, Laffey, Pham, Fan, Brochard, Esteban, van Haren, Larsson, McAuley, Ranieri, Wrigge, Slutsky.
*Drafting of the manuscript:* Bellani, Laffey, Pham, Fan, Ranieri, Thompson.
*Critical revision of the manuscript for important intellectual content:* Bellani, Laffey, Pham, Fan, Brochard, Esteban, Gattinoni, van Haren, Larsson, McAuley, Ranieri, Rubenfeld, Wrigge, Slutsky, Pesenti.
*Statistical analysis:* Bellani, Laffey, Pham, Fan, Ranieri.
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*Administrative, technical, or material support:* Bellani, Laffey, Fan, Esteban, McAuley, Ranieri, Thompson, Slutsky.
*Study supervision:* Bellani, Laffey, Brochard, Esteban, Gattinoni, van Haren, Larsson, Ranieri, Slutsky, Pesenti.

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**Group Information:** LUNG SAFE Investigators and the ESICM Trials Group are listed in the Supplement.

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